

Syllabus

ALICE CORPORATION PTY. LTD. *v.* CLS BANK
INTERNATIONAL ET AL.CERTIORARI TO THE UNITED STATES COURT OF APPEALS FOR
THE FEDERAL CIRCUIT

No. 13–298. Argued March 31, 2014—Decided June 19, 2014

Petitioner Alice Corporation is the assignee of several patents that disclose a scheme for mitigating “settlement risk,” *i. e.*, the risk that only one party to an agreed-upon financial exchange will satisfy its obligation. In particular, the patent claims are designed to facilitate the exchange of financial obligations between two parties by using a computer system as a third-party intermediary. The patents in suit claim (1) a method for exchanging financial obligations, (2) a computer system configured to carry out the method for exchanging obligations, and (3) a computer-readable medium containing program code for performing the method of exchanging obligations.

Respondents (together, CLS Bank), who operate a global network that facilitates currency transactions, filed suit against petitioner, arguing that the patent claims at issue are invalid, unenforceable, or not infringed. Petitioner counterclaimed, alleging infringement. After *Bilski v. Kappos*, 561 U.S. 593, was decided, the District Court held that all of the claims were ineligible for patent protection under 35 U.S.C. § 101 because they are directed to an abstract idea. The en banc Federal Circuit affirmed.

Held: Because the claims are drawn to a patent-ineligible abstract idea, they are not patent eligible under § 101. Pp. 216–227.

(a) The Court has long held that § 101, which defines the subject matter eligible for patent protection, contains an implicit exception for “[l]aws of nature, natural phenomena, and abstract ideas.” *Association for Molecular Pathology v. Myriad Genetics, Inc.*, 569 U.S. 576, 589. In applying the § 101 exception, this Court must distinguish patents that claim the “‘buildin[g] block[s]’” of human ingenuity, which are ineligible for patent protection, from those that integrate the building blocks into something more, see *Mayo Collaborative Services v. Prometheus Laboratories, Inc.*, 566 U.S. 66, 89, thereby “transform[ing]” them into a patent-eligible invention, *id.*, at 72. Pp. 216–217.

(b) Using this framework, the Court must first determine whether the claims at issue are directed to a patent-ineligible concept. 566 U.S., at 77. If so, the Court then asks whether the claim’s elements, consid-

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ered both individually and “as an ordered combination,” “transform the nature of the claim” into a patent-eligible application. *Id.*, at 79, 78. Pp. 217–227.

(1) The claims at issue are directed to a patent-ineligible concept: the abstract idea of intermediated settlement. Under “the longstanding rule that [a]n idea of itself is not patentable,” *Gottschalk v. Benson*, 409 U. S. 63, 67, this Court has found ineligible patent claims involving an algorithm for converting binary-coded decimal numerals into pure binary form, *id.*, at 71–72; a mathematical formula for computing “alarm limits” in a catalytic conversion process, *Parker v. Flook*, 437 U. S. 584, 594–595; and, most recently, a method for hedging against the financial risk of price fluctuations, *Bilski*, 561 U. S., at 599. It follows from these cases, and *Bilski* in particular, that the claims at issue are directed to an abstract idea. On their face, they are drawn to the concept of intermediated settlement, *i. e.*, the use of a third party to mitigate settlement risk. Like the risk hedging in *Bilski*, the concept of intermediated settlement is “‘a fundamental economic practice long prevalent in our system of commerce,’” *ibid.*, and the use of a third-party intermediary (or “clearing house”) is a building block of the modern economy. Thus, intermediated settlement, like hedging, is an “abstract idea” beyond § 101’s scope. Pp. 218–221.

(2) Turning to the second step of *Mayo*’s framework: The method claims, which merely require generic computer implementation, fail to transform that abstract idea into a patent-eligible invention. Pp. 221–226.

(i) “[S]imply appending conventional steps, specified at a high level of generality,” to a method already “well known in the art” is not “enough” to supply the “‘inventive concept’” needed to make this transformation. *Mayo, supra*, at 82, 79, 77, 72. The introduction of a computer into the claims does not alter the analysis. Neither stating an abstract idea “while adding the words ‘apply it,’” *Mayo, supra*, at 72, nor limiting the use of an abstract idea “‘to a particular technological environment,’” *Bilski, supra*, at 610–611, is enough for patent eligibility. Stating an abstract idea while adding the words “apply it with a computer” simply combines those two steps, with the same deficient result. Wholly generic computer implementation is not generally the sort of “additional featur[e]” that provides any “practical assurance that the process is more than a drafting effort designed to monopolize the [abstract idea] itself.” *Mayo, supra*, at 77. Pp. 221–224.

(ii) Here, the representative method claim does no more than simply instruct the practitioner to implement the abstract idea of intermediated settlement on a generic computer. Taking the claim elements

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separately, the function performed by the computer at each step—creating and maintaining “shadow” accounts, obtaining data, adjusting account balances, and issuing automated instructions—is “[p]urely ‘conventional.’” *Mayo*, 566 U.S., at 79. Considered “as an ordered combination,” these computer components “ad[d] nothing . . . that is not already present when the steps are considered separately.” *Ibid.* Viewed as a whole, these method claims simply recite the concept of intermediated settlement as performed by a generic computer. They do not, for example, purport to improve the functioning of the computer itself or effect an improvement in any other technology or technical field. An instruction to apply the abstract idea of intermediated settlement using some unspecified, generic computer is not “enough” to transform the abstract idea into a patent-eligible invention. *Id.*, at 77. Pp. 224–226.

(3) Because petitioner’s system and media claims add nothing of substance to the underlying abstract idea, they too are patent ineligible under § 101. Petitioner conceded below that its media claims rise or fall with its method claims. And the system claims are no different in substance from the method claims. The method claims recite the abstract idea implemented on a generic computer; the system claims recite a handful of generic computer components configured to implement the same idea. This Court has long “warn[ed] . . . against” interpreting § 101 “in ways that make patent eligibility ‘depend simply on the draftsman’s art.’” *Mayo, supra*, at 72. Holding that the system claims are patent eligible would have exactly that result. Pp. 226–227.

717 F. 3d 1269, affirmed.

THOMAS, J., delivered the opinion for a unanimous Court. SOTOMAYOR, J., filed a concurring opinion, in which GINSBURG and BREYER, JJ., joined, *post*, p. 227.

Carter G. Phillips argued the cause for petitioner. With him on the briefs were *Jeffrey P. Kushan*, *Constantine L. Trela, Jr.*, *Tacy F. Flint*, *Adam L. Perlman*, and *Robert E. Sokohl*.

Mark A. Perry argued the cause for respondents. With him on the brief were *Helgi C. Walker* and *Brian M. Buroker*.

Solicitor General Verrilli argued the cause for the United States as *amicus curiae* urging affirmance. With him on

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the brief were *Assistant Attorney General Delery, Deputy Solicitor General Stewart, Ginger D. Anders, Mark R. Freeman, and Scott C. Weidenfeller*.*

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JUSTICE THOMAS delivered the opinion of the Court.

The patents at issue in this case disclose a computer-implemented scheme for mitigating “settlement risk” (*i. e.*, the risk that only one party to a financial transaction will pay what it owes) by using a third-party intermediary. The question presented is whether these claims are patent eligible under 35 U. S. C. § 101, or are instead drawn to a patent-ineligible abstract idea. We hold that the claims at issue are drawn to the abstract idea of intermediated settlement, and that merely requiring generic computer implementation fails to transform that abstract idea into a patent-eligible invention. We therefore affirm the judgment of the United States Court of Appeals for the Federal Circuit.

I

A

Petitioner Alice Corporation is the assignee of several patents that disclose schemes to manage certain forms of financial risk.¹ According to the specification largely shared by the patents, the invention “enabl[es] the management of risk relating to specified, yet unknown, future events.” App. 248. The specification further explains that the “invention relates to methods and apparatus, including electrical com-

for the New York Intellectual Property Law Association by *Anthony F. Lo Cicero*, *Charles R. Macedo*, *Joseph M. Casino*, *Michael J. Kasdan*, and *Bruce D. Abramson*; for Proove Biosciences, Inc., by *Catherine Meshkin* and *Patrick R. Delaney*; for Retailers by *Peter J. Brann* and *Stacy O. Stitham*; for SHFL Entertainment, Inc., by *Adrian M. Pruetz*, *Erica J. Van Loon*, *Rex Hwang*, and *Charles C. Koole*; for Sigram Schindler Beteiligungsgesellschaft mbH by *Chidambaram S. Iyer*; for Ronald M. Benrey by *Robert R. Sachs* and *Daniel R. Brownstone*; for Dale R. Cook by *Mr. Cook, pro se*; for Robin Feldman et al. by *Mr. Feldman, pro se*; for Brian R. Galvin by *Roy Rainey*; for Lee Hollaar et al. by *Peter K. Trzyna, pro se*; for James B. Lampert et al. by *Mr. Lampert, pro se*; and for Paul R. Michel by *Charles Hieken* and *John A. Dragseth*.

¹The patents at issue are United States Patent Nos. 5,970,479 (the ‘479 patent), 6,912,510, 7,149,720, and 7,725,375.

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puters and data processing systems applied to financial matters and risk management.” *Id.*, at 243.

The claims at issue relate to a computerized scheme for mitigating “settlement risk”—*i. e.*, the risk that only one party to an agreed-upon financial exchange will satisfy its obligation. In particular, the claims are designed to facilitate the exchange of financial obligations between two parties by using a computer system as a third-party intermediary. *Id.*, at 383–384.² The intermediary creates “shadow” credit and debit records (*i. e.*, account ledgers) that mirror the balances in the parties’ real-world accounts at “exchange institutions” (*e. g.*, banks). The intermediary updates the shadow records in real time as transactions are entered, allowing “only those transactions for which the parties’ updated shadow records indicate sufficient resources to satisfy their mutual obligations.” 717 F. 3d 1269, 1285 (CA Fed.

²The parties agree that claim 33 of the ’479 patent is representative of the method claims. Claim 33 recites:

“A method of exchanging obligations as between parties, each party holding a credit record and a debit record with an exchange institution, the credit records and debit records for exchange of predetermined obligations, the method comprising the steps of:

“(a) creating a shadow credit record and a shadow debit record for each stakeholder party to be held independently by a supervisory institution from the exchange institutions;

“(b) obtaining from each exchange institution a start-of-day balance for each shadow credit record and shadow debit record;

“(c) for every transaction resulting in an exchange obligation, the supervisory institution adjusting each respective party’s shadow credit record or shadow debit record, allowing only those transactions that do not result in the value of the shadow debit record being less than the value of the shadow credit record at any time, each said adjustment taking place in chronological order, and

“(d) at the end-of-day, the supervisory institution instructing on[e] of the exchange institutions to exchange credits or debits to the credit record and debit record of the respective parties in accordance with the adjustments of the said permitted transactions, the credits and debits being irrevocable, time invariant obligations placed on the exchange institutions.” App. 383–384.

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2013) (Lourie, J., concurring). At the end of the day, the intermediary instructs the relevant financial institutions to carry out the “permitted” transactions in accordance with the updated shadow records, *ibid.*, thus mitigating the risk that only one party will perform the agreed-upon exchange.

In sum, the patents in suit claim (1) the foregoing method for exchanging obligations (the method claims), (2) a computer system configured to carry out the method for exchanging obligations (the system claims), and (3) a computer-readable medium containing program code for performing the method of exchanging obligations (the media claims). All of the claims are implemented using a computer; the system and media claims expressly recite a computer, and the parties have stipulated that the method claims require a computer as well.

B

Respondents CLS Bank International and CLS Services Ltd. (together, CLS Bank) operate a global network that facilitates currency transactions. In 2007, CLS Bank filed suit against petitioner, seeking a declaratory judgment that the claims at issue are invalid, unenforceable, or not infringed. Petitioner counterclaimed, alleging infringement. Following this Court’s decision in *Bilski v. Kappos*, 561 U. S. 593 (2010), the parties filed cross-motions for summary judgment on whether the asserted claims are eligible for patent protection under 35 U. S. C. § 101. The District Court held that all of the claims are patent ineligible because they are directed to the abstract idea of “employing a neutral intermediary to facilitate simultaneous exchange of obligations in order to minimize risk.” 768 F. Supp. 2d 221, 252 (DC 2011).

A divided panel of the United States Court of Appeals for the Federal Circuit reversed, holding that it was not “manifestly evident” that petitioner’s claims are directed to an abstract idea. 685 F. 3d 1341, 1352, 1356 (2012). The Federal Circuit granted rehearing en banc, vacated the panel opinion, and affirmed the judgment of the District Court in a one-

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paragraph *per curiam* opinion. 717 F. 3d, at 1273. Seven of the ten participating judges agreed that petitioner’s method and media claims are patent ineligible. See *id.*, at 1274 (Lourie, J., concurring); *id.*, at 1312–1313 (Rader, C. J., concurring in part and dissenting in part). With respect to petitioner’s system claims, the en banc Federal Circuit affirmed the District Court’s judgment by an equally divided vote. *Id.*, at 1273.

Writing for a five-member plurality, Judge Lourie concluded that all of the claims at issue are patent ineligible. In the plurality’s view, under this Court’s decision in *Mayo Collaborative Services v. Prometheus Laboratories, Inc.*, 566 U. S. 66 (2012), a court must first “identif[y] the abstract idea represented in the claim,” and then determine “whether the balance of the claim adds ‘significantly more.’” 717 F. 3d, at 1286. The plurality concluded that petitioner’s claims “draw on the abstract idea of reducing settlement risk by effecting trades through a third-party intermediary,” and that the use of a computer to maintain, adjust, and reconcile shadow accounts added nothing of substance to that abstract idea. *Ibid.*

Chief Judge Rader concurred in part and dissented in part. In a part of the opinion joined only by Judge Moore, Chief Judge Rader agreed with the plurality that petitioner’s method and media claims are drawn to an abstract idea. *Id.*, at 1312–1313. In a part of the opinion joined by Judges Linn, Moore, and O’Malley, Chief Judge Rader would have held that the system claims are patent eligible because they involve computer “hardware” that is “specifically programmed to solve a complex problem.” *Id.*, at 1307. Judge Moore wrote a separate opinion dissenting in part, arguing that the system claims are patent eligible. *Id.*, at 1313–1314. Judge Newman filed an opinion concurring in part and dissenting in part, arguing that all of petitioner’s claims are patent eligible. *Id.*, at 1327. Judges Linn and O’Malley filed a separate dissenting opinion reaching that same conclusion. *Ibid.*

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We granted certiorari, 571 U. S. 1090 (2013), and now affirm.

II

Section 101 of the Patent Act defines the subject matter eligible for patent protection. It provides:

“Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.” 35 U. S. C. § 101.

“We have long held that this provision contains an important implicit exception: Laws of nature, natural phenomena, and abstract ideas are not patentable.” *Association for Molecular Pathology v. Myriad Genetics, Inc.*, 569 U. S. 576, 589 (2013) (internal quotation marks and brackets omitted). We have interpreted § 101 and its predecessors in light of this exception for more than 150 years. *Bilski*, *supra*, at 601–602; see also *O’Reilly v. Morse*, 15 How. 62, 112–120 (1854); *Le Roy v. Tatham*, 14 How. 156, 174–175 (1853).

We have described the concern that drives this exclusionary principle as one of pre-emption. See, e. g., *Bilski*, *supra*, at 611–612 (upholding the patent “would pre-empt use of this approach in all fields, and would effectively grant a monopoly over an abstract idea”). Laws of nature, natural phenomena, and abstract ideas are ““the basic tools of scientific and technological work.”” *Myriad*, *supra*, at 589. “[M]onopolization of those tools through the grant of a patent might tend to impede innovation more than it would tend to promote it,” thereby thwarting the primary object of the patent laws. *Mayo*, *supra*, at 71; see U. S. Const., Art. I, § 8, cl. 8 (Congress “shall have Power . . . To promote the Progress of Science and useful Arts”). We have “repeatedly emphasized this . . . concern that patent law not inhibit further discovery by improperly tying up the future use of” these building blocks of human ingenuity. *Mayo*, *supra*, at 85 (citing *Morse*, *supra*, at 113).

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At the same time, we tread carefully in construing this exclusionary principle lest it swallow all of patent law. *Mayo*, 566 U. S., at 71. At some level, “all inventions . . . embody, use, reflect, rest upon, or apply laws of nature, natural phenomena, or abstract ideas.” *Ibid.* Thus, an invention is not rendered ineligible for patent simply because it involves an abstract concept. See *Diamond v. Diehr*, 450 U. S. 175, 187 (1981). “[A]pplication[s]” of such concepts “‘to a new and useful end,’” we have said, remain eligible for patent protection. *Gottschalk v. Benson*, 409 U. S. 63, 67 (1972).

Accordingly, in applying the § 101 exception, we must distinguish between patents that claim the “‘buildin[g] block[s]’” of human ingenuity and those that integrate the building blocks into something more, *Mayo*, 566 U. S., at 89, thereby “transform[ing]” them into a patent-eligible invention, *id.*, at 72. The former “would risk disproportionately tying up the use of the underlying” ideas, *id.*, at 73, and are therefore ineligible for patent protection. The latter pose no comparable risk of pre-emption, and therefore remain eligible for the monopoly granted under our patent laws.

III

In *Mayo*, we set forth a framework for distinguishing patents that claim laws of nature, natural phenomena, and abstract ideas from those that claim patent-eligible applications of those concepts. First, we determine whether the claims at issue are directed to one of those patent-ineligible concepts. *Id.*, at 77. If so, we then ask, “[w]hat else is there in the claims before us?” *Id.*, at 78. To answer that question, we consider the elements of each claim both individually and “as an ordered combination” to determine whether the additional elements “transform the nature of the claim” into a patent-eligible application. *Id.*, at 79, 78. We have described step two of this analysis as a search for an “‘inventive concept’”—*i. e.*, an element or combination of elements

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that is “sufficient to ensure that the patent in practice amounts to significantly more than a patent upon the [ineligible concept] itself.” *Id.*, at 72–73.³

A

We must first determine whether the claims at issue are directed to a patent-ineligible concept. We conclude that they are: These claims are drawn to the abstract idea of intermediated settlement.

The “abstract ideas” category embodies “the longstanding rule that ‘[a]n idea of itself is not patentable.’” *Benson*, *supra*, at 67 (quoting *Rubber-Tip Pencil Co. v. Howard*, 20 Wall. 498, 507 (1874)); see also *Le Roy*, *supra*, at 175 (“A principle, in the abstract, is a fundamental truth; an original cause; a motive; these cannot be patented, as no one can claim in either of them an exclusive right”). In *Benson*, for example, this Court rejected as ineligible patent claims involving an algorithm for converting binary-coded decimal numerals into pure binary form, holding that the claimed patent was “in practical effect . . . a patent on the algorithm itself.” 409 U.S., at 71–72. And in *Parker v. Flook*, 437 U.S. 584, 594–595 (1978), we held that a mathematical formula for computing “alarm limits” in a catalytic conversion process was also a patent-ineligible abstract idea.

We most recently addressed the category of abstract ideas in *Bilski*, 561 U.S. 593. The claims at issue in *Bilski* described a method for hedging against the financial risk of price fluctuations. Claim 1 recited a series of steps for hedging risk, including: (1) initiating a series of financial transactions between providers and consumers of a commodity; (2)

³Because the approach we made explicit in *Mayo* considers all claim elements, both individually and in combination, it is consistent with the general rule that patent claims “must be considered as a whole.” *Diamond v. Diehr*, 450 U.S. 175, 188 (1981); see *Parker v. Flook*, 437 U.S. 584, 594 (1978) (“Our approach . . . is . . . not at all inconsistent with the view that a patent claim must be considered as a whole”).

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identifying market participants that have a counterrisk for the same commodity; and (3) initiating a series of transactions between those market participants and the commodity provider to balance the risk position of the first series of consumer transactions. *Id.*, at 599. Claim 4 “pu[t] the concept articulated in claim 1 into a simple mathematical formula.” *Ibid.* The remaining claims were drawn to examples of hedging in commodities and energy markets.

“[A]ll Members of the Court agree[d]” that the patent at issue in *Bilski* claimed an “abstract idea.” *Id.*, at 609; see also *id.*, at 619 (Stevens, J., concurring in judgment). Specifically, the claims described “the basic concept of hedging, or protecting against risk.” *Id.*, at 611. The Court explained that “[h]edging is a fundamental economic practice long prevalent in our system of commerce and taught in any introductory finance class.” *Ibid.* “The concept of hedging” as recited by the claims in suit was therefore a patent-ineligible “abstract idea, just like the algorithms at issue in *Benson* and *Flook*.” *Ibid.*

It follows from our prior cases, and *Bilski* in particular, that the claims at issue here are directed to an abstract idea. Petitioner’s claims involve a method of exchanging financial obligations between two parties using a third-party intermediary to mitigate settlement risk. The intermediary creates and updates “shadow” records to reflect the value of each party’s actual accounts held at “exchange institutions,” thereby permitting only those transactions for which the parties have sufficient resources. At the end of each day, the intermediary issues irrevocable instructions to the exchange institutions to carry out the permitted transactions.

On their face, the claims before us are drawn to the concept of intermediated settlement, *i. e.*, the use of a third party to mitigate settlement risk. Like the risk hedging in *Bilski*, the concept of intermediated settlement is “‘a fundamental economic practice long prevalent in our system of commerce.’” *Ibid.*; see, *e. g.*, Emery, Speculation on the

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Stock and Produce Exchanges of the United States, in 7 Studies in History, Economics and Public Law 283, 346–356 (1896) (discussing the use of a “clearing-house” as an intermediary to reduce settlement risk). The use of a third-party intermediary (or “clearinghouse”) is also a building block of the modern economy. See, *e. g.*, Yadav, The Problematic Case of Clearinghouses in Complex Markets, 101 Geo. L. J. 387, 406–412 (2013); J. Hull, Risk Management and Financial Institutions 103–104 (3d ed. 2012). Thus, intermediated settlement, like hedging, is an “abstract idea” beyond the scope of § 101.

Petitioner acknowledges that its claims describe intermediated settlement, see Brief for Petitioner 4, but rejects the conclusion that its claims recite an “abstract idea.” Drawing on the presence of mathematical formulas in some of our abstract-ideas precedents, petitioner contends that the abstract-ideas category is confined to “preexisting, fundamental truth[s]” that “‘exis[t] in principle apart from any human action.’” *Id.*, at 23, 26 (quoting *Mayo*, 566 U.S., at 77).

Bilski belies petitioner’s assertion. The concept of risk hedging we identified as an abstract idea in that case cannot be described as a “preexisting, fundamental truth.” The patent in *Bilski* simply involved a “series of steps instructing how to hedge risk.” 561 U.S., at 599. Although hedging is a longstanding commercial practice, *ibid.*, it is a method of organizing human activity, not a “truth” about the natural world “‘that has always existed,’” Brief for Petitioner 22 (quoting *Flook*, *supra*, at 593, n. 15). One of the claims in *Bilski* reduced hedging to a mathematical formula, but the Court did not assign any special significance to that fact, much less the sort of talismanic significance petitioner claims. Instead, the Court grounded its conclusion that all of the claims at issue were abstract ideas in the understanding that risk hedging was a “‘fundamental economic practice.’” 561 U.S., at 611.

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In any event, we need not labor to delimit the precise contours of the “abstract ideas” category in this case. It is enough to recognize that there is no meaningful distinction between the concept of risk hedging in *Bilski* and the concept of intermediated settlement at issue here. Both are squarely within the realm of “abstract ideas” as we have used that term.

B

Because the claims at issue are directed to the abstract idea of intermediated settlement, we turn to the second step in *Mayo*’s framework. We conclude that the method claims, which merely require generic computer implementation, fail to transform that abstract idea into a patent-eligible invention.

1

At *Mayo* step two, we must examine the elements of the claim to determine whether it contains an “‘inventive concept’” sufficient to “transform” the claimed abstract idea into a patent-eligible application. 566 U. S., at 72, 80. A claim that recites an abstract idea must include “additional features” to ensure “that the [claim] is more than a drafting effort designed to monopolize the [abstract idea].” *Id.*, at 77. *Mayo* made clear that transformation into a patent-eligible application requires “more than simply stat[ing] the [abstract idea] while adding the words ‘apply it.’” *Id.*, at 72.

Mayo itself is instructive. The patents at issue in *Mayo* claimed a method for measuring metabolites in the bloodstream in order to calibrate the appropriate dosage of thio-purine drugs in the treatment of autoimmune diseases. *Id.*, at 73–75. The respondent in that case contended that the claimed method was a patent-eligible application of natural laws that describe the relationship between the concentration of certain metabolites and the likelihood that the drug dosage will be harmful or ineffective. But methods for determining metabolite levels were already “well known in the

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art,” and the process at issue amounted to “nothing significantly more than an instruction to doctors to apply the applicable laws when treating their patients.” *Id.*, at 79. “[S]imply appending conventional steps, specified at a high level of generality,” was not “enough” to supply an “‘inventive concept.’” *Id.*, at 82, 77, 72.

The introduction of a computer into the claims does not alter the analysis at *Mayo* step two. In *Benson*, for example, we considered a patent that claimed an algorithm implemented on “a general-purpose digital computer.” 409 U. S., at 64. Because the algorithm was an abstract idea, see *supra*, at 218, the claim had to supply a “‘new and useful’” application of the idea in order to be patent eligible. 409 U. S., at 67. But the computer implementation did not supply the necessary inventive concept; the process could be “carried out in existing computers long in use.” *Ibid.* We accordingly “held that simply implementing a mathematical principle on a physical machine, namely a computer, [i]s not a patentable application of that principle.” *Mayo, supra*, at 84 (citing *Benson, supra*, at 64).

Flook is to the same effect. There, we examined a computerized method for using a mathematical formula to adjust alarm limits for certain operating conditions (*e. g.*, temperature and pressure) that could signal inefficiency or danger in a catalytic conversion process. 437 U. S., at 585–586. Once again, the formula itself was an abstract idea, see *supra*, at 218, and the computer implementation was purely conventional. 437 U. S., at 594 (noting that the “use of computers for ‘automatic monitoring-alarming’” was “well known”). In holding that the process was patent ineligible, we rejected the argument that “implement[ing] a principle in some specific fashion” will “automatically fal[l] within the patentable subject matter of § 101.” *Id.*, at 593. Thus, “*Flook* stands for the proposition that the prohibition against patenting abstract ideas cannot be circumvented by attempting to limit the use of [the idea] to a particular technological environment.”

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Bilski, 561 U.S., at 610–611 (internal quotation marks omitted).

In *Diehr*, 450 U.S. 175, by contrast, we held that a computer-implemented process for curing rubber was patent eligible, but not because it involved a computer. The claim employed a “well-known” mathematical equation, but it used that equation in a process designed to solve a technological problem in “conventional industry practice.” *Id.*, at 177, 178. The invention in *Diehr* used a “thermocouple” to record constant temperature measurements inside the rubber mold—something “the industry ha[d] not been able to obtain.” *Id.*, at 178, and n. 3. The temperature measurements were then fed into a computer, which repeatedly recalculated the remaining cure time by using the mathematical equation. *Id.*, at 178–179. These additional steps, we recently explained, “transformed the process into an inventive application of the formula.” *Mayo, supra*, at 81. In other words, the claims in *Diehr* were patent eligible because they improved an existing technological process, not because they were implemented on a computer.

These cases demonstrate that the mere recitation of a generic computer cannot transform a patent-ineligible abstract idea into a patent-eligible invention. Stating an abstract idea “while adding the words ‘apply it’” is not enough for patent eligibility. *Mayo, supra*, at 72. Nor is limiting the use of an abstract idea “‘to a particular technological environment.’” *Bilski, supra*, at 610–611. Stating an abstract idea while adding the words “apply it with a computer” simply combines those two steps, with the same deficient result. Thus, if a patent’s recitation of a computer amounts to a mere instruction to “implemen[t]” an abstract idea “on . . . a computer,” *Mayo, supra*, at 84, that addition cannot impart patent eligibility. This conclusion accords with the pre-emption concern that undergirds our § 101 jurisprudence. Given the ubiquity of computers, see 717 F.3d, at 1286 (Lourie, J., concurring), wholly generic computer im-

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plementation is not generally the sort of “additional featur[e]” that provides any “practical assurance that the process is more than a drafting effort designed to monopolize the [abstract idea] itself.” *Mayo, supra*, at 77.

The fact that a computer “necessarily exist[s] in the physical, rather than purely conceptual, realm,” Brief for Petitioner 39, is beside the point. There is no dispute that a computer is a tangible system (in § 101 terms, a “machine”), or that many computer-implemented claims are formally addressed to patent-eligible subject matter. But if that were the end of the § 101 inquiry, an applicant could claim any principle of the physical or social sciences by reciting a computer system configured to implement the relevant concept. Such a result would make the determination of patent eligibility “depend simply on the draftsman’s art,” *Flook, supra*, at 593, thereby eviscerating the rule that “[l]aws of nature, natural phenomena, and abstract ideas are not patentable,” *Myriad*, 569 U. S., at 589.

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The representative method claim in this case recites the following steps: (1) “creating” shadow records for each counterparty to a transaction; (2) “obtaining” start-of-day balances based on the parties’ real-world accounts at exchange institutions; (3) “adjusting” the shadow records as transactions are entered, allowing only those transactions for which the parties have sufficient resources; and (4) issuing irrevocable end-of-day instructions to the exchange institutions to carry out the permitted transactions. See n. 2, *supra*. Petitioner principally contends that the claims are patent eligible because these steps “require a substantial and meaningful role for the computer.” Brief for Petitioner 48. As stipulated, the claimed method requires the use of a computer to create electronic records, track multiple transactions, and issue simultaneous instructions; in other words, “[t]he computer is itself the intermediary.” *Ibid.* (emphasis deleted).

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In light of the foregoing, see *supra*, at 222, the relevant question is whether the claims here do more than simply instruct the practitioner to implement the abstract idea of intermediated settlement on a generic computer. They do not.

Taking the claim elements separately, the function performed by the computer at each step of the process is “[p]urely conventional.” *Mayo*, 566 U. S., at 79 (internal quotation marks omitted). Using a computer to create and maintain “shadow” accounts amounts to electronic recordkeeping—one of the most basic functions of a computer. See, e. g., *Benson*, 409 U. S., at 65 (noting that a computer “operates . . . upon both new and previously stored data”). The same is true with respect to the use of a computer to obtain data, adjust account balances, and issue automated instructions; all of these computer functions are “well-understood, routine, conventional activit[ies]” previously known to the industry. *Mayo*, 566 U. S., at 73. In short, each step does no more than require a generic computer to perform generic computer functions.

Considered “as an ordered combination,” the computer components of petitioner’s method “ad[d] nothing . . . that is not already present when the steps are considered separately.” *Id.*, at 79. Viewed as a whole, petitioner’s method claims simply recite the concept of intermediated settlement as performed by a generic computer. See 717 F. 3d, at 1286 (Lourie, J., concurring) (noting that the representative method claim “lacks *any* express language to define the computer’s participation”). The method claims do not, for example, purport to improve the functioning of the computer itself. See *ibid.* (“There is no specific or limiting recitation of . . . improved computer technology . . .”); Brief for United States as *Amicus Curiae* 28–30. Nor do they effect an improvement in any other technology or technical field. See, e. g., *Diehr*, *supra*, at 177–178. Instead, the claims at issue amount to “nothing significantly more” than

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an instruction to apply the abstract idea of intermediated settlement using some unspecified, generic computer. *Mayo*, 566 U.S., at 79. Under our precedents, that is not “enough” to transform an abstract idea into a patent-eligible invention. *Id.*, at 77.

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Petitioner’s claims to a computer system and a computer-readable medium fail for substantially the same reasons. Petitioner conceded below that its media claims rise or fall with its method claims. En Banc Response Brief for Defendant-Appellant in No. 11–1301 (CA Fed.), p. 50, n. 3. As to its system claims, petitioner emphasizes that those claims recite “specific hardware” configured to perform “specific computerized functions.” Brief for Petitioner 53. But what petitioner characterizes as specific hardware—a “data processing system” with a “communications controller” and “data storage unit,” for example, see App. 954, 958, 1257—is purely functional and generic. Nearly every computer will include a “communications controller” and “data storage unit” capable of performing the basic calculation, storage, and transmission functions required by the method claims. See 717 F. 3d, at 1290 (Lourie, J., concurring). As a result, none of the hardware recited by the system claims “offers a meaningful limitation beyond generally linking ‘the use of the [method] to a particular technological environment,’ that is, implementation via computers.” *Id.*, at 1291 (quoting *Bilski*, 561 U.S., at 610–611).

Put another way, the system claims are no different from the method claims in substance. The method claims recite the abstract idea implemented on a generic computer; the system claims recite a handful of generic computer components configured to implement the same idea. This Court has long “warn[ed] . . . against” interpreting § 101 “in ways that make patent eligibility ‘depend simply on the draftsman’s art.’” *Mayo, supra*, at 72 (quoting *Flook*, 437 U.S., at 593); see *id.*, at 590 (“The concept of patentable subject

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matter under § 101 is not ‘like a nose of wax which may be turned and twisted in any direction . . . ’”). Holding that the system claims are patent eligible would have exactly that result.

Because petitioner’s system and media claims add nothing of substance to the underlying abstract idea, we hold that they too are patent ineligible under § 101.

* * *

For the foregoing reasons, the judgment of the Court of Appeals for the Federal Circuit is affirmed.

It is so ordered.

JUSTICE SOTOMAYOR, with whom JUSTICE GINSBURG and JUSTICE BREYER join, concurring.

I adhere to the view that any “claim that merely describes a method of doing business does not qualify as a ‘process’ under § 101.” *Bilski v. Kappos*, 561 U. S. 593, 614 (2010) (Stevens, J., concurring in judgment); see also *In re Bilski*, 545 F. 3d 943, 972 (CA Fed. 2008) (Dyk, J., concurring) (“There is no suggestion in any of th[e] early [English] consideration of process patents that processes for organizing human activity were or ever had been patentable”). As in *Bilski*, however, I further believe that the method claims at issue are drawn to an abstract idea. Cf. 561 U. S., at 619 (opinion of Stevens, J.). I therefore join the opinion of the Court.